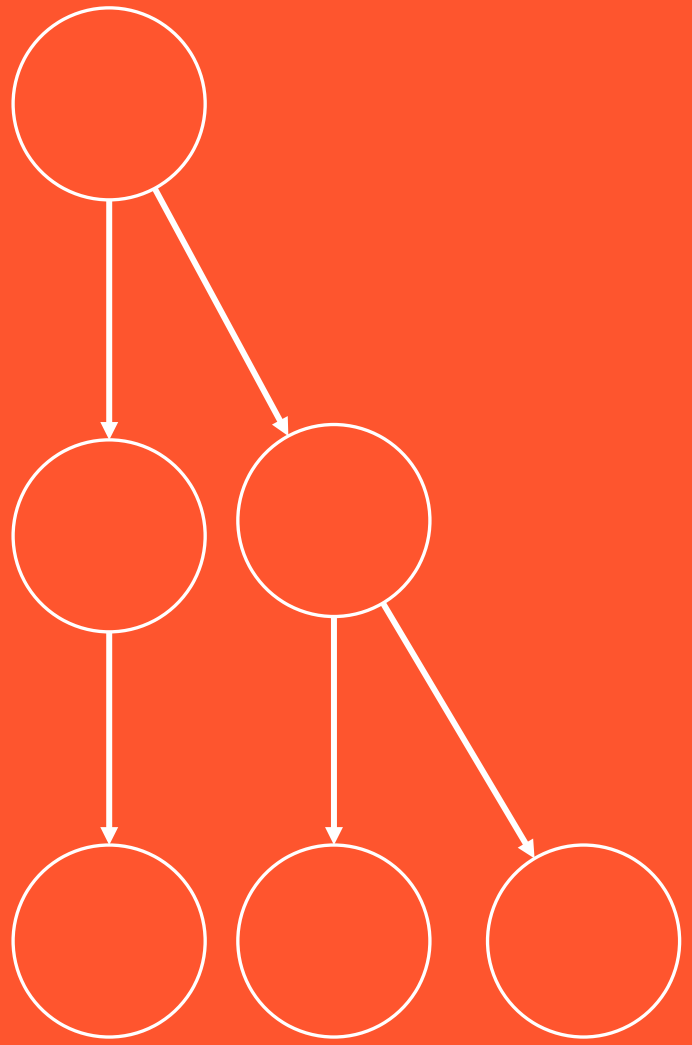


BCOG 100

Learning and Plasticity

Lecture 5.3 — The Biology of Neural Learning

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: The Four Kinds of Learning

In your group, three minutes.

In the last lecture, we discussed four kinds of learning algorithms used by living and artificial systems.

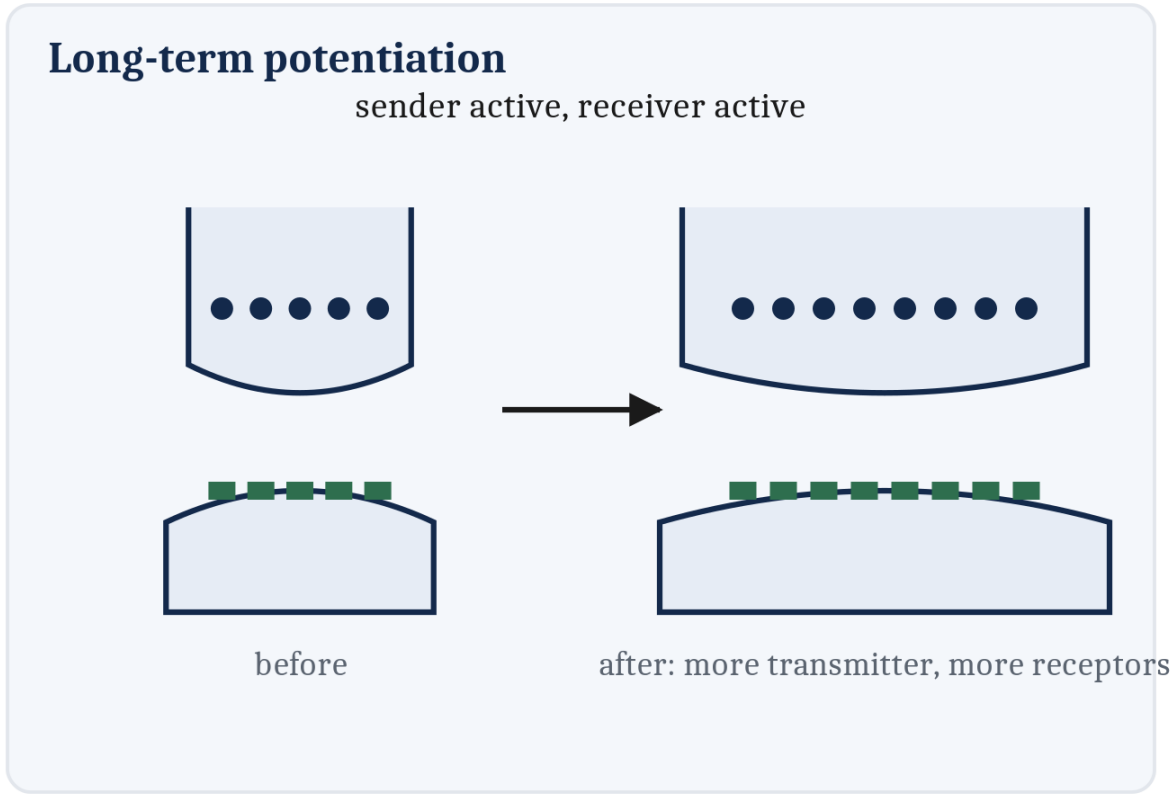
What are they called?

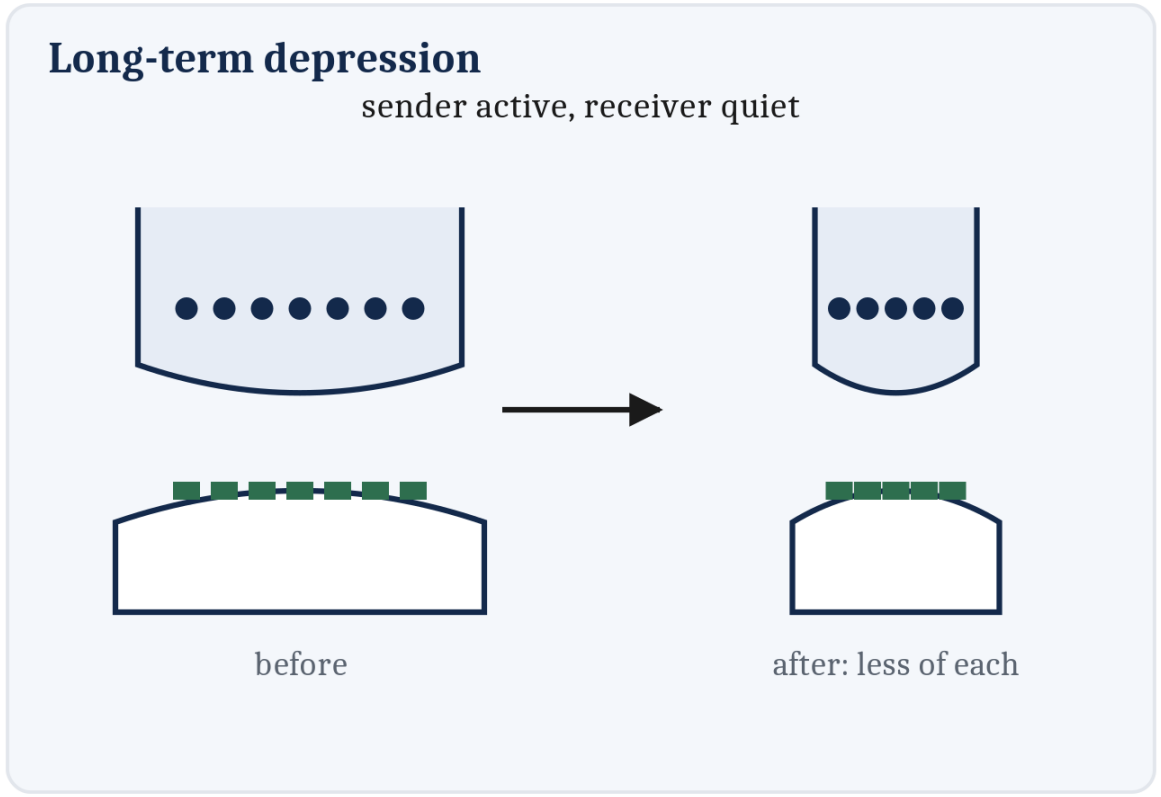
How are they different?

Which are easiest and hardest for a learner to use and implement?

Strengthening a Synapse, and Weakening One

Fire together repeatedly and the synapse strengthens: **long-term potentiation**. Sender active while the receiver stays quiet, and it weakens: **long-term depression** — which is what keeps the rule from running away.





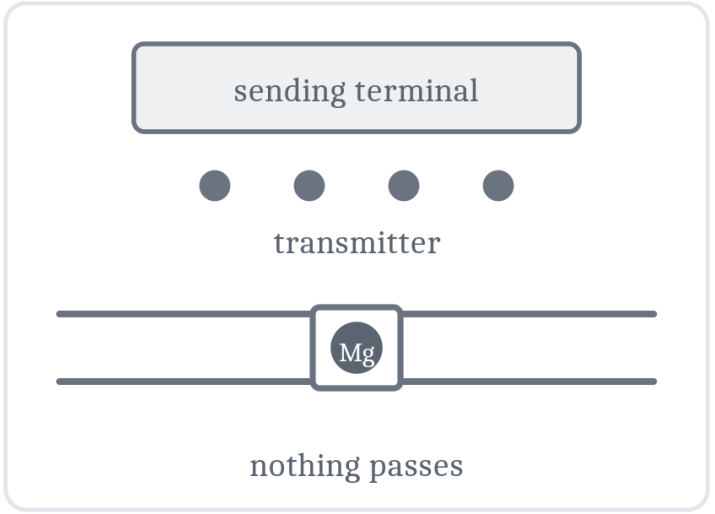
The pair together is what lets a weight move both ways.

A synapse that could only strengthen would be a one-way ratchet.

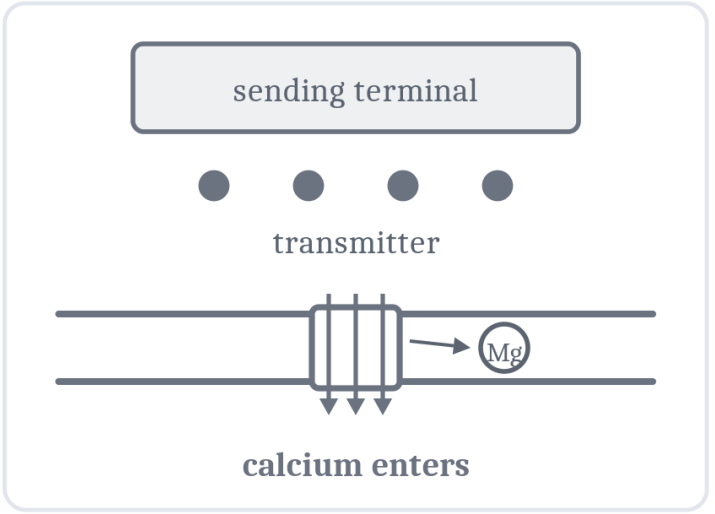
The NMDA Receptor Is an AND Gate

The **NMDA receptor** opens only when transmitter is bound *and* the receiving cell is already depolarized. That is **coincidence detection**, and here the name is the mechanism.

A truth table, built out of chemistry



cell not depolarized: blocked



cell depolarized: open

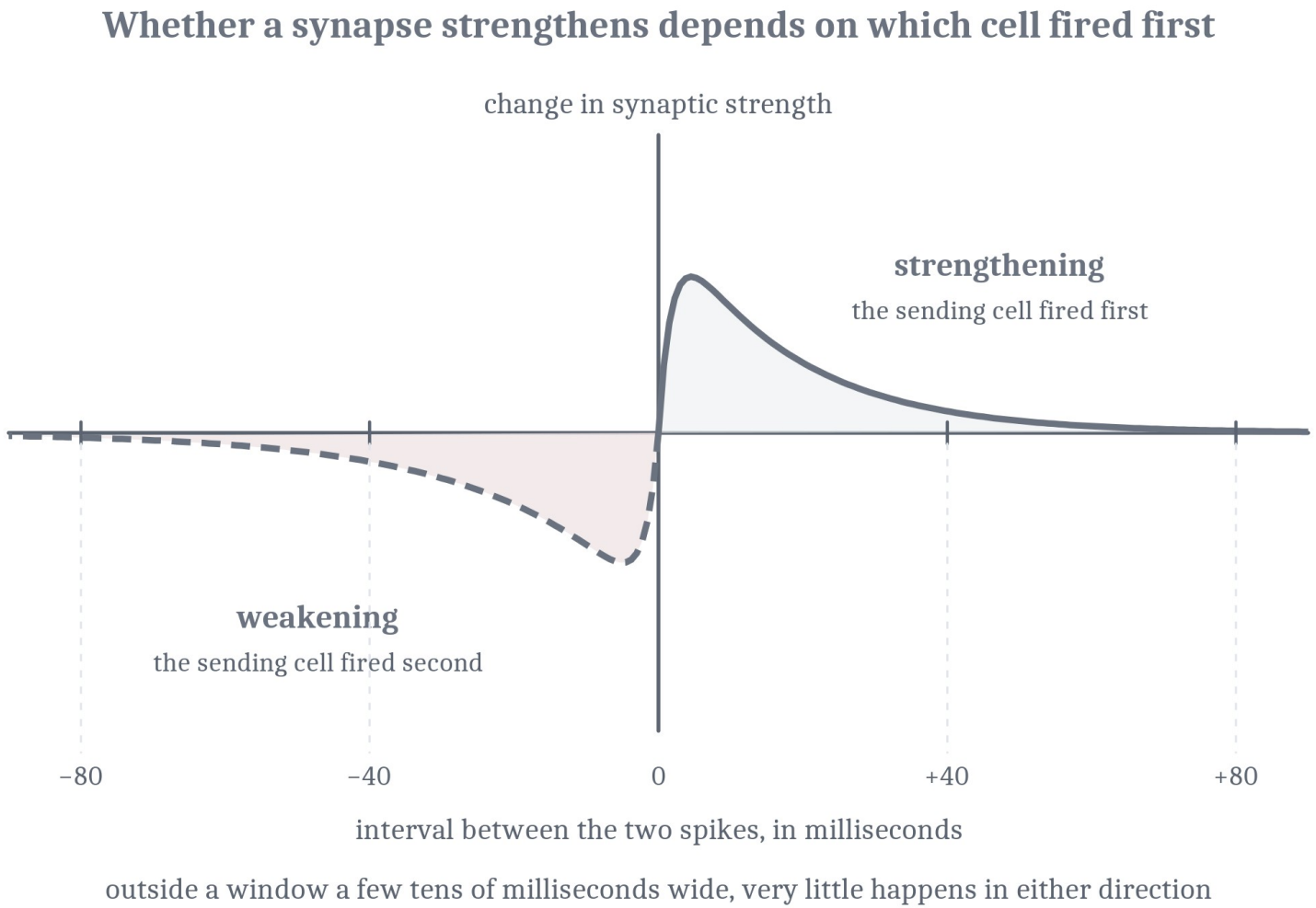
transmitter	depolarized	calcium
no	no	no
yes	no	no
no	yes	no
yes	yes	**yes**

AND

either condition alone does nothing; only both together open the channel

Timing, and Which One Came First

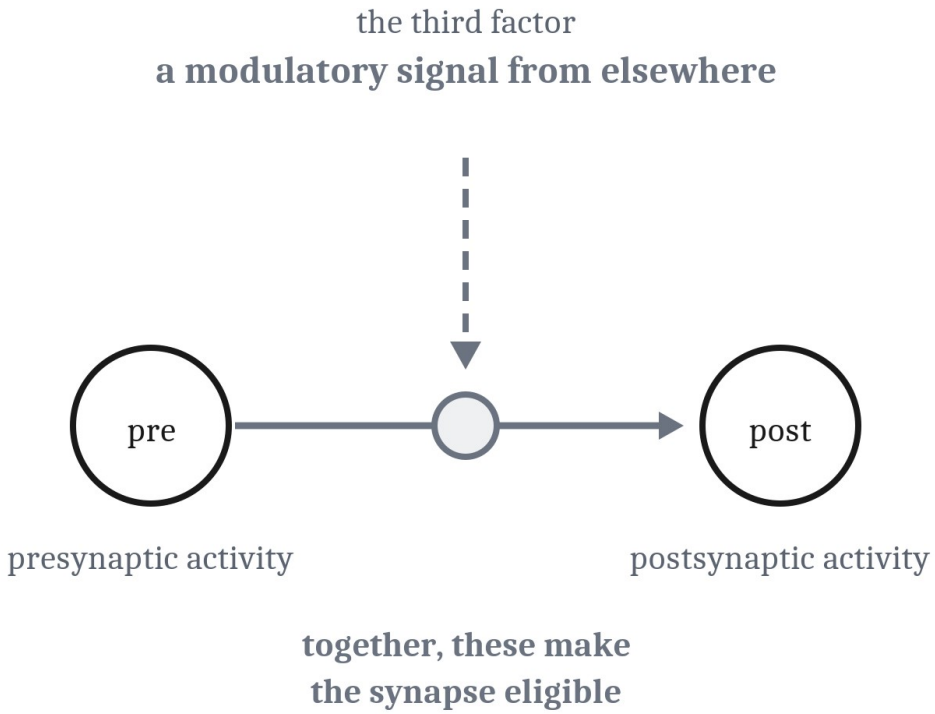
Sender fires just before the receiver: the synapse strengthens. Just after: it weakens. The window is a few tens of milliseconds — **spike-timing-dependent plasticity**.

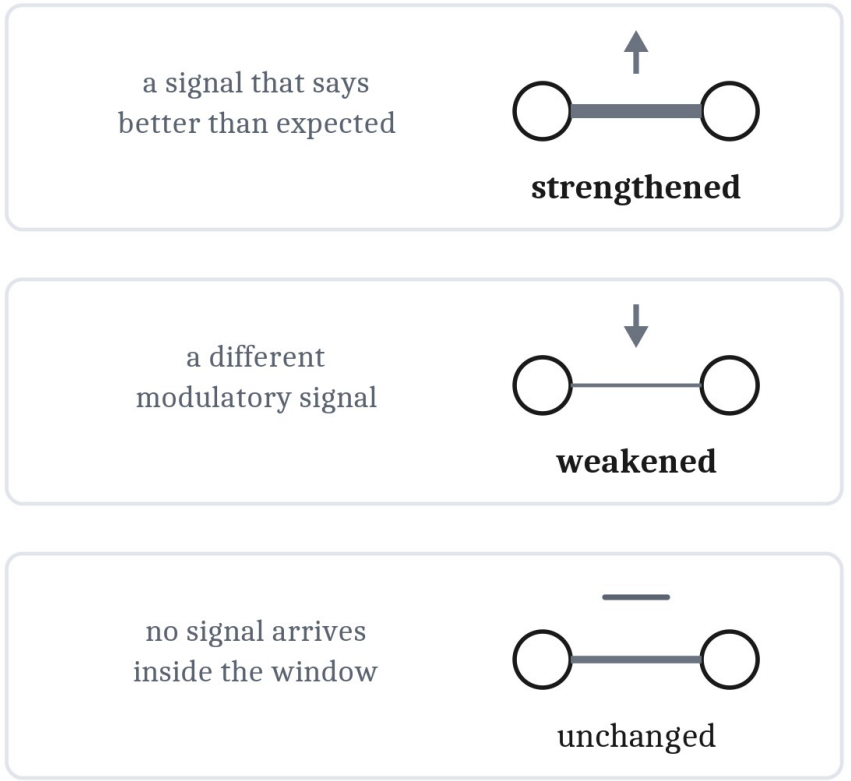


The Three-Factor Rule

Presynaptic activity and postsynaptic activity decide which synapses are eligible. A third signal, broadcast from elsewhere, decides whether they change — and in **which direction**.

Two cells make a synapse eligible. A third signal decides what happens.

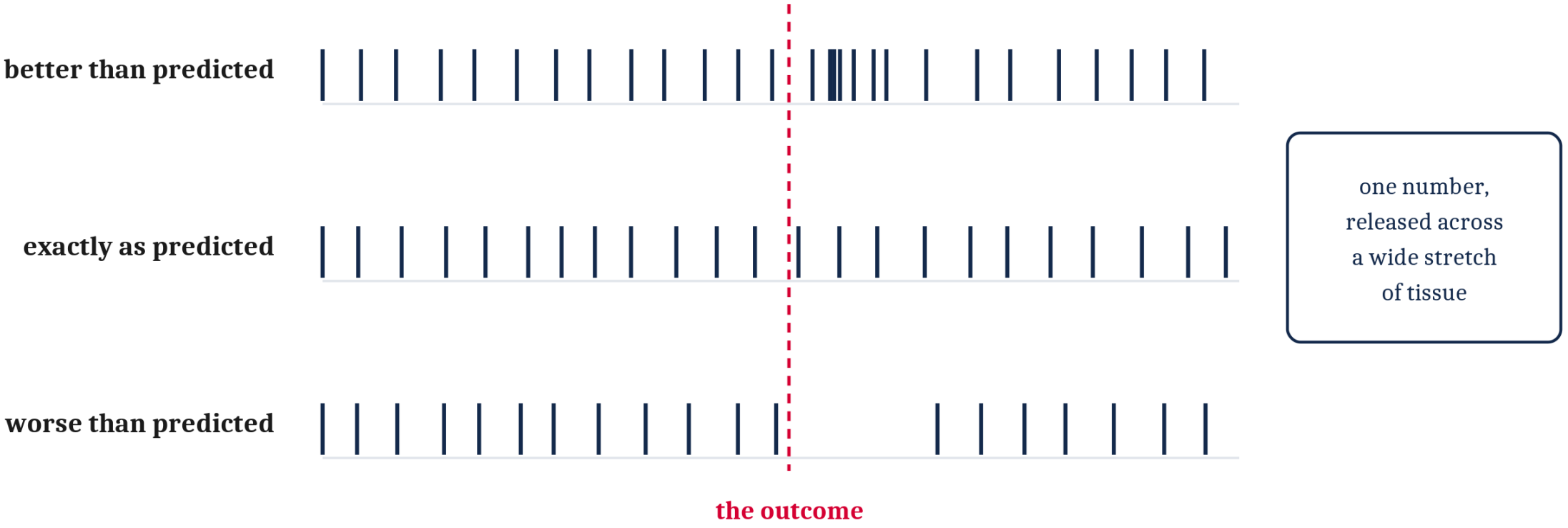




the third factor can reverse the direction of the change, not merely scale its size

Dopamine Reports Better and Worse Than Expected

Above baseline when an outcome beats the prediction, below it when the outcome falls short, unchanged when it arrives as predicted: the **reward prediction error**, broadcast.



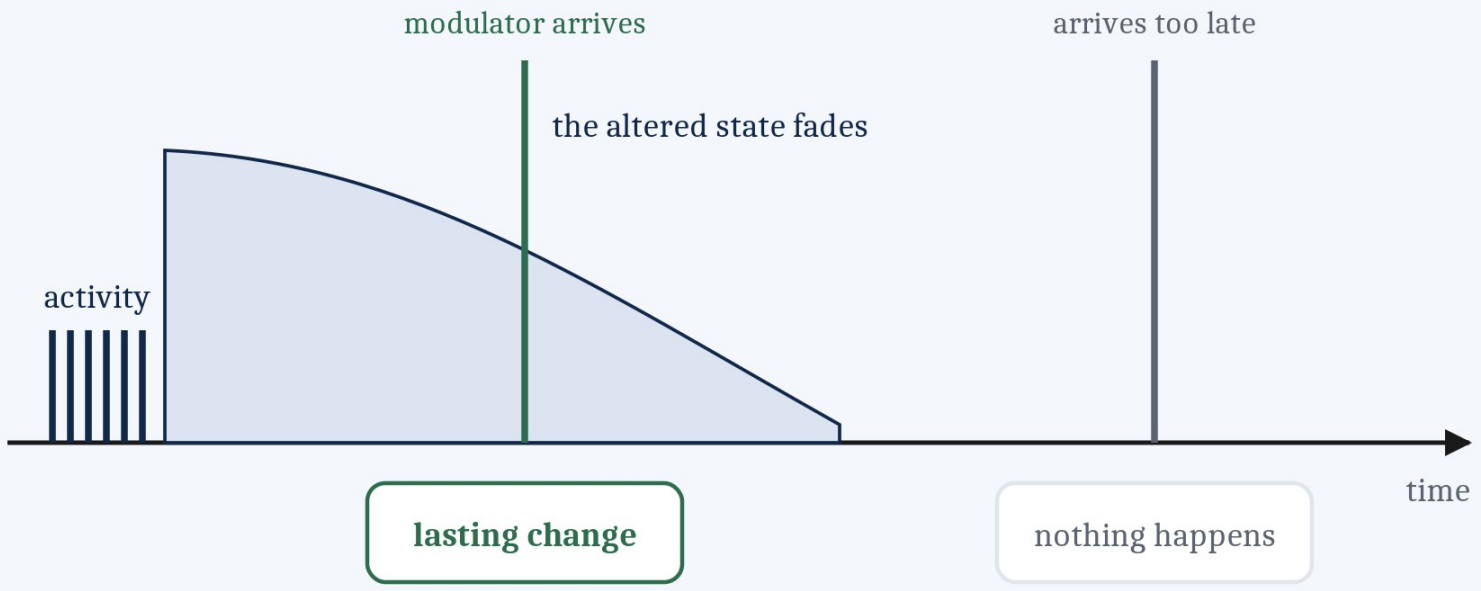
Above baseline, at baseline, below baseline: the reward prediction error.

Schematic. The recordings were made in vertebrates, largely primates.

The Trace Made Physical, and the Other Signal From Elsewhere

A synapse that was just active is left in an altered state that fades over seconds. A modulatory signal inside that window makes the change last; after it, nothing happens.

At a synapse, over a few seconds



The other signal



one climbing fiber

one cell, one instruction

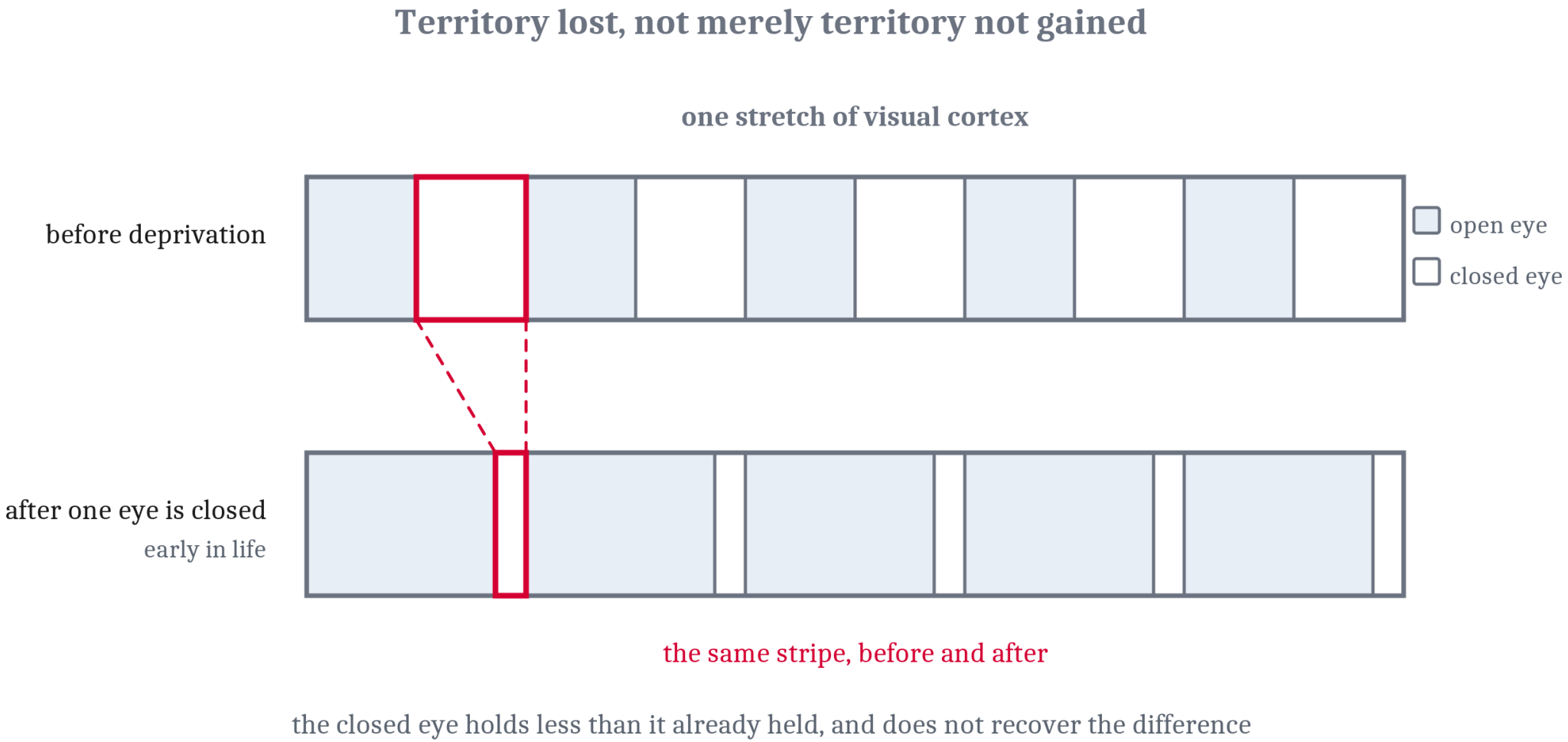
the cerebellum: Module 6

The thing the algorithm needed is the sort of thing a synapse can be.

That does not show the algorithm is correct.

Learning Has a Schedule

Connections are made in profusion (**synaptogenesis**), then cut back by activity (**pruning**). A **critical period** is a window when experience has an outsized, lasting effect on wiring.



Name One, Then Say How It Closes

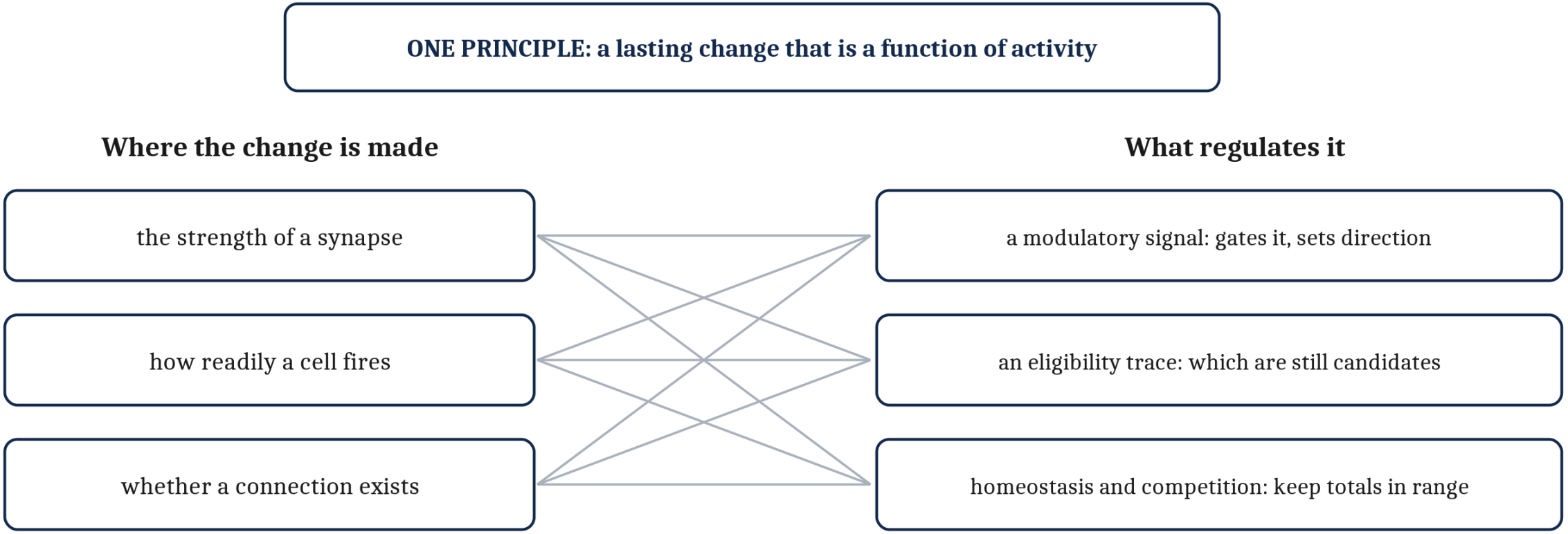
In your group, four minutes.

Name something that looks like it has a **critical period** — a capacity where early experience matters in a way later experience cannot match. Anything except the two cases from the reading.

Then: **what would have to be true of the brain for that window to close?**

One Principle, Several Loci, Several Regulators

Three places to change: synaptic strength, **intrinsic excitability**, whether a connection exists. Three regulators: a modulatory signal, an eligibility trace, **homeostatic plasticity** with competition.



Every mechanism in this lecture is one of each.

A Synapse Changed and Stayed Changed. Is That a Memory?

A **disposition** is committed when it is laid down. A **record** is kept without its use being settled, and what it is for is decided at retrieval. Learning produces dispositions.

	A disposition	A record
Committed at encoding	Yes — what the change will do is fixed by the change	No — what it is for is settled later
A separate retrieval step	Not needed; the change acts whenever the situation recurs	Required, and it is where the use gets decided
What it can report	What tends to happen	What happened, on a particular occasion
Where it is taken up	Module 5	Module 10

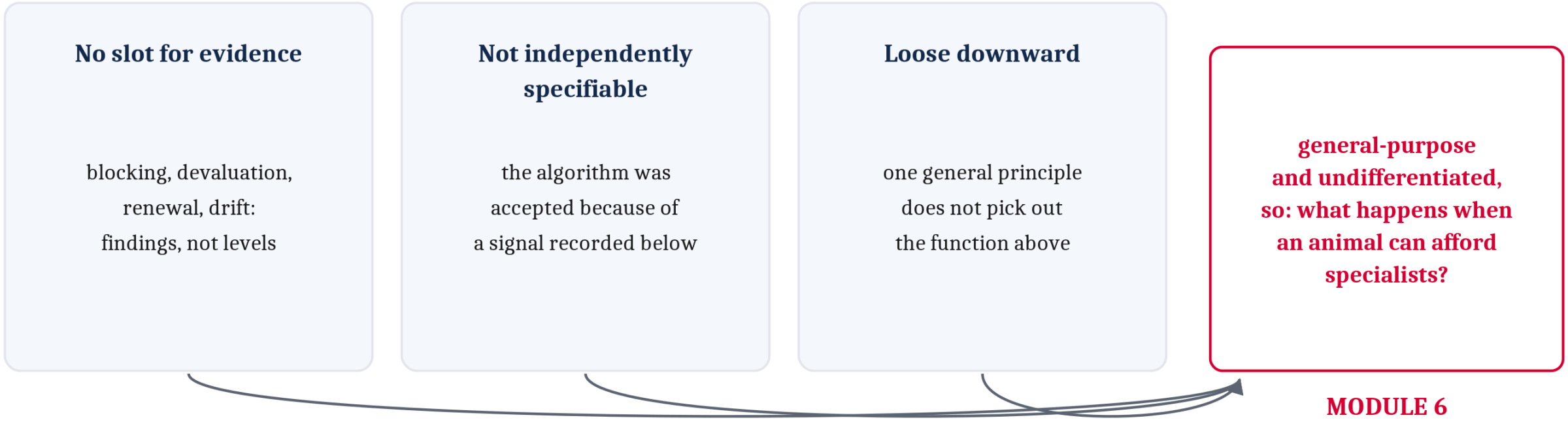
What Separates the Four Kinds of Learning

Not what they change: all four change a weight. What separates them is how much of the world has to reach that weight before it moves, and who supplies the answer. One synapse, given the right third factor, implements all of it.

Vocabulary about goals and errors earns its place exactly where something in the mechanism plays the part the word names.

Three Places Marr's Levels Strained, and What Comes Next

The levels have no slot for evidence. They were not independently specifiable here. And the implementation did not pick out one function. A principle reusable everywhere is specialized for nothing.



A framework is a tool with a shape; knowing where it strains is part of using it.

Still waiting: Module 7 (a single unit's limits), Module 10 (records), Module 12 (models).

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4	The NMDA receptor beside the truth table for AND	fig-nmda-and.png	CC BY 4.0	Jon Willits — drawn for the Module 5 reading · own work
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